

Active Directory Domain Services (AD DS) Implementation Report

Enterprise SOC Lab – Configuration, Findings & Key Facts

1. Objective

The objective of implementing **Active Directory Domain Services (AD DS)** was to establish a centralized identity and authentication service for the enterprise SOC lab. The Domain Controller provides centralized management of users, computers, authentication, DNS, and future integrations with Linux-based security infrastructure.

2. Enterprise Network Architecture

The AD deployment follows a segmented enterprise network architecture.

DMZ Network

Network: 192.168.100.0/24

Purpose

- Public-facing services
- Internet-accessible systems

Hosts

Host	IP Address
Web Server	192.168.100.10
Honeypot	192.168.100.100
Future AWS Web Server Planned	

Internal Security Network

Network: 192.168.150.0/24

Purpose

- Security infrastructure
- Active Directory
- DNS
- SIEM
- Threat Intelligence

Hosts

Host	IP Address
MISP	192.168.150.10
SOC-Core	192.168.150.20
IDS	192.168.150.30
Domain Controller	192.168.150.40

Edge Gateway (AWS)

Ubuntu-based VPN Gateway performing:

- WireGuard VPN Hub
- Routing
- NAT
- Internet Edge Gateway
- Planned nftables firewall

Current deployment uses a single Elastic IP, with a future plan to allocate a second Elastic IP for dedicated web hosting.

3. Domain Controller Configuration

Operating System

- Windows Server 2022

Hostname

- KPK-DC01

Domain

- kpk.local

Internal IP

- 192.168.150.40

Installed Roles

- Active Directory Domain Services (AD DS)
- DNS Server
- Global Catalog

Verification

The following were successfully verified:

- Domain Controller operational
- Global Catalog enabled
- All FSMO roles assigned
- LDAP service available
- Kerberos service available

PowerShell verification:

Get-ADDomainController

Get-ADDomain

4. DNS Infrastructure

DNS was deployed as an integrated Active Directory service.

Configured Zones

Forward Lookup Zones

- kpk.local
- _msdcs.kpk.local

The Domain Controller acts as the authoritative DNS server for the enterprise environment.

Primary DNS Server

192.168.150.40

5. Forward Lookup Records

Static DNS host (A) records were created for all enterprise systems.

Hostname FQDN		IP Address
KPK-DC01	kpk-dc01.kpk.local	192.168.150.40
misp	misp.kpk.local	192.168.150.10
soc-core	soc-core.kpk.local	192.168.150.20
ids	ids.kpk.local	192.168.150.30
web	web.kpk.local	192.168.100.10
honeypot	honeypot.kpk.local	192.168.100.100

DNS resolution was verified using:

Resolve-DnsName

6. Active Directory Service Discovery

The following SRV records were successfully verified:

LDAP

_ldap._tcp.dc._msdcs.kpk.local

Kerberos

_kerberos._tcp.dc._msdcs.kpk.local

Both services correctly advertise:

KPK-DC01

↓

192.168.150.40

This confirms successful Active Directory service discovery for future Windows and Linux domain members.

7. Major Issue Encountered

Problem

LDAP and Kerberos initially advertised the incorrect IP address:

192.168.74.176

instead of

192.168.150.40

Later, another incorrect address,

10.35.119.1

was also advertised.

Root Cause

The Domain Controller was configured with multiple active network interfaces:

- WireGuard Interface
- VMware Network Interface
- USB Tethering Interface

Netlogon automatically registered all eligible interfaces in Active Directory DNS, causing incorrect service advertisements.

Resolution

The following corrective actions were performed:

- Updated the Domain Controller DNS A record.
- Disabled DNS registration on non-enterprise interfaces.
- Flushed DNS cache.
- Re-registered DNS records.
- Restarted the Netlogon service.
- Forced Active Directory DNS registration.

Commands executed:

```
ipconfig /flushdns
```

```
ipconfig /registerdns
```

```
Restart-Service Netlogon
```

```
nltest /dsregdns
```

Final Result

LDAP and Kerberos now advertise only:

192.168.150.40

ensuring correct Active Directory service discovery.

8. Validation Performed

Hostname Resolution

Verified using:

```
Resolve-DnsName web.kpk.local
```

```
Resolve-DnsName ids.kpk.local
```

```
Resolve-DnsName misp.kpk.local
```

```
Resolve-DnsName soc-core.kpk.local
```

```
Resolve-DnsName honeypot.kpk.local
```

All hostnames resolved successfully.

Active Directory Service Discovery

Verified using:

Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.kpk.local

Resolve-DnsName -Type SRV _kerberos._tcp.dc._msdcs.kpk.local

Both returned the correct Domain Controller.

DNS Registration

Verified using:

nslookup KPK-DC01.kpk.local

Returned:

192.168.150.40

Domain Controller Health

Verified using:

Get-ADDomainController

Get-ADDomain

All services reported healthy.

9. Security Design Decisions

Network Segmentation

DMZ

Contains:

- Web Server
- Honeypot

Purpose:

Public-facing services isolated from internal infrastructure.

Internal Security Network

Contains:

- Domain Controller
- SOC-Core
- IDS
- MISP

Purpose:

Hosts enterprise identity, monitoring, and threat intelligence infrastructure.

Honeypot Isolation

The Honeypot was intentionally kept outside Active Directory.

Reason

Since honeypots are designed to attract malicious activity, isolating them from the enterprise identity infrastructure minimizes risk and reflects enterprise security practices.

10. Current AD Implementation Status

Successfully completed:

- Active Directory Domain Services deployment
 - Windows Server 2022 Domain Controller configuration
 - DNS Server installation
 - Global Catalog configuration
 - Domain creation (kpk.local)
 - Active Directory integrated DNS configuration
 - Forward Lookup Zone configuration
 - Enterprise DNS host record creation
 - LDAP SRV record validation
 - Kerberos SRV record validation
 - Resolution of multi-homed Domain Controller DNS registration issues
 - Validation of enterprise hostname resolution
 - Verification of Active Directory service discovery
-

11. Key Technical Findings

- Active Directory is highly dependent on DNS; incorrect DNS registration prevents proper domain service discovery.
- Multi-homed Domain Controllers may register multiple IP addresses, causing LDAP and Kerberos to advertise incorrect interfaces.
- Only the enterprise-facing interface should be allowed to register in Active Directory DNS.
- LDAP and Kerberos rely on SRV records for automatic Domain Controller discovery, making SRV validation an essential deployment step.
- Forward Lookup Zones and static host records simplify enterprise hostname resolution and future domain integration.
- Proper network segmentation between DMZ and internal infrastructure reduces attack surface and aligns with enterprise security architecture.
- Maintaining the Domain Controller on a dedicated internal network ensures reliable authentication and centralized identity management.

12. Final Outcome

A fully operational **Windows Server 2022 Active Directory Domain Services** environment was successfully deployed as the centralized identity platform for the enterprise SOC lab. The deployment included Active Directory-integrated DNS, enterprise hostname resolution, validation of LDAP and Kerberos service discovery, and resolution of a critical multi-homed DNS registration issue. The final configuration provides a stable and standards-aligned foundation for centralized authentication and enterprise security infrastructure integration.

Phase 1 – Active Directory Domain Services (AD DS) Implementation

Enterprise SOC Lab

This phase covers the deployment and validation of **Active Directory Domain Services (AD DS)** and **DNS** on the Windows Server Domain Controller. It establishes the identity infrastructure required for centralized authentication and name resolution within the enterprise SOC environment.

Step 1: Configure Windows Server

Assign a **static IP** to the Domain Controller.

Hostname : KPK-DC01

IP : 192.168.150.40

Subnet : 255.255.255.0

Gateway : VPN Gateway

DNS : 127.0.0.1

Verify configuration:

hostname

ipconfig /all

Step 2: Install AD DS and DNS

Open

Server Manager

→ Add Roles and Features

Install

- Active Directory Domain Services
- DNS Server

After installation, promote the server to a Domain Controller.

Step 3: Promote Server to Domain Controller

Server Manager

Notifications

↓

Promote this server to a domain controller

Configuration

Deployment Operation

Add a new forest

Root Domain

kpk.local

Set

- DSRM Password
- Forest Functional Level
- Domain Functional Level

Install and reboot.

Step 4: Verify Domain Controller

(Get-CimInstance Win32_ComputerSystem).Domain

Expected

kpk.local

Verify AD DS

Get-WindowsFeature AD-Domain-Services

Verify DNS

Get-WindowsFeature DNS

Verify Domain Controller

Get-ADDomainController

Verify Domain

Get-ADDomain

Expected

- Domain Controller Active
- Global Catalog Enabled
- FSMO Roles Assigned

Step 5: Verify DNS Installation

List DNS Zones

Get-DnsServerZone

Expected

kpk.local

_msdcs.kpk.local

Verify Forwarders

Get-DnsServerForwarder

Step 6: Create Enterprise DNS Records

Open

Server Manager

↓

Tools

↓

DNS

Navigate

Forward Lookup Zones

↓

kpk.local

Right Click

New Host (A or AAAA)

Create the following records.

Host	IP
KPK-DC01	192.168.150.40
misp	192.168.150.10
soc-core	192.168.150.20
ids	192.168.150.30
web	192.168.100.10
honeypot	192.168.100.100

Step 7: Verify Hostname Resolution

Run

```
Resolve-DnsName misp.kpk.local
```

```
Resolve-DnsName soc-core.kpk.local
```

```
Resolve-DnsName ids.kpk.local
```

```
Resolve-DnsName web.kpk.local
```

```
Resolve-DnsName honeypot.kpk.local
```

Expected

All hostnames resolve to their configured IP addresses.

Step 8: Verify Active Directory DNS Service Discovery

Verify LDAP

```
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.kpk.local
```

Verify Kerberos

```
Resolve-DnsName -Type SRV _kerberos._tcp.dc._msdcs.kpk.local
```

Expected

```
KPK-DC01.kpk.local
```

↓

192.168.150.40

Step 9: Verify DNS Registration

Resolve-DnsName KPK-DC01.kpk.local

nslookup KPK-DC01.kpk.local 127.0.0.1

Expected

192.168.150.40

Step 10: Refresh Domain Controller DNS Registration

After network changes, refresh DNS registration.

ipconfig /flushdns

ipconfig /registerdns

Restart-Service Netlogon

nltest /dsregdns

Verify again

Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.kpk.local

Resolve-DnsName -Type SRV _kerberos._tcp.dc._msdcs.kpk.local

Expected

Additional Section

192.168.150.40

Step 11: Validate Domain Controller Health

Get-ADDomainController

Get-ADDomain

hostname

ipconfig /all

Verify

- Domain Name
- Hostname
- IP Address
- Global Catalog
- FSMO Roles
- LDAP
- Kerberos
- DNS

Phase 1 Deliverables

- Windows Server 2022 configured as Domain Controller.
- New Active Directory forest and domain (kpk.local) created.
- DNS Server installed and integrated with AD DS.
- Forward Lookup Zones configured.
- Enterprise DNS host (A) records created.
- LDAP and Kerberos SRV records validated.
- Domain Controller advertising the correct internal enterprise IP (192.168.150.40).
- Hostname resolution verified for all enterprise systems.
- Active Directory and DNS services confirmed healthy and ready for integration with the SOC infrastructure.

This concludes **Phase 1: Active Directory Domain Services Deployment and DNS Configuration**, providing the foundation for subsequent Linux domain joins, centralized authentication, and security service integration.

Phase 2 – Active Directory DNS Integration & Enterprise DNS Configuration

Enterprise SOC Lab

This phase focuses on integrating the enterprise infrastructure with the Domain Controller by configuring DNS records, validating Active Directory service discovery, and preparing the environment for centralized authentication.

Step 1: Configure Enterprise Network Addressing

Final network architecture used for AD integration.

DMZ

192.168.100.0/24

Host	IP
Web Server	192.168.100.10
Honeypot	192.168.100.100

Internal Security Network

192.168.150.0/24

Host	IP
MISP	192.168.150.10
SOC-Core	192.168.150.20
IDS	192.168.150.30
Domain Controller	192.168.150.40

Step 2: Configure Forward Lookup Records

Open

Server Manager

→ Tools

→ DNS

Navigate

Forward Lookup Zones

→ kpk.local

Create **Host (A)** records.

Host	FQDN	IP
KPK-DC01	kpk-dc01.kpk.local	192.168.150.40
misp	misp.kpk.local	192.168.150.10
soc-core	soc-core.kpk.local	192.168.150.20
ids	ids.kpk.local	192.168.150.30
web	web.kpk.local	192.168.100.10
honeypot	honeypot.kpk.local	192.168.100.100

Step 3: Verify Forward DNS Resolution

Run

```
Resolve-DnsName misp.kpk.local
```

```
Resolve-DnsName soc-core.kpk.local
```

```
Resolve-DnsName ids.kpk.local
```

```
Resolve-DnsName web.kpk.local
```

```
Resolve-DnsName honeypot.kpk.local
```

Expected

- All hostnames resolve to their configured IP addresses.
-

Step 4: Verify Active Directory Service Discovery

Verify LDAP

```
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.kpk.local
```

Verify Kerberos

```
Resolve-DnsName -Type SRV _kerberos._tcp.dc._msdcs.kpk.local
```

Expected

```
KPK-DC01.kpk.local
```

↓

192.168.150.40

Step 5: Validate Domain Controller Registration

Resolve-DnsName KPK-DC01.kpk.local

nslookup KPK-DC01.kpk.local 127.0.0.1

Expected

192.168.150.40

Step 6: Refresh Active Directory DNS Registration

After changing network configuration or DNS records:

ipconfig /flushdns

ipconfig /registerdns

Restart-Service Netlogon

nltest /dsregdns

Verify again

Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.kpk.local

Resolve-DnsName -Type SRV _kerberos._tcp.dc._msdcs.kpk.local

Expected

Additional Section

192.168.150.40

Step 7: Configure Linux Clients to Use AD DNS

All enterprise Linux servers should use the Domain Controller as their primary DNS server.

VM	DNS Server
SOC-Core	192.168.150.40
MISP	192.168.150.40
IDS	192.168.150.40
Web Server	192.168.150.40
Honeypot	192.168.150.40 (optional if isolated)

Example (Netplan):

nameservers:

addresses:

- 192.168.150.40

Apply:

sudo netplan apply

Step 8: Verify DNS from Linux

Install DNS utilities if required:

sudo apt update

sudo apt install dnsutils -y

Verify hostname resolution:

dig kpk.local

dig kpk-dc01.kpk.local

dig misp.kpk.local

dig soc-core.kpk.local

dig ids.kpk.local

dig web.kpk.local

Verify Active Directory SRV records:

```
dig -t SRV _ldap._tcp.dc._msdcs.kpk.local
```

```
dig -t SRV _kerberos._tcp.dc._msdcs.kpk.local
```

Or use:

```
nslookup kpk-dc01.kpk.local 192.168.150.40
```

Step 9: Validate Connectivity

From each Linux VM:

```
ping kpk-dc01.kpk.local
```

```
ping misp.kpk.local
```

```
ping soc-core.kpk.local
```

Confirm that name resolution works through the Domain Controller instead of using IP addresses directly.

Phase 2 Deliverables

- Enterprise DNS namespace (kpk.local) established.
- DNS A records created for all infrastructure hosts.
- Domain Controller successfully advertising LDAP and Kerberos services.
- DNS registration refreshed and validated.
- Linux infrastructure configured to use the AD-integrated DNS server (192.168.150.40).
- Forward DNS resolution verified from both Windows and Linux.
- Enterprise infrastructure ready for Active Directory domain joining and centralized authentication.

Phase 3 – Linux Integration with Active Directory

Enterprise SOC Lab

This phase integrates the Linux-based security infrastructure with Active Directory, enabling centralized authentication, domain-based identity management, and enterprise DNS resolution.

Objective

Join Linux servers to the Active Directory domain (kpk.local) and enable centralized authentication using AD user accounts.

Target Systems

- SOC-Core
- MISP
- IDS

Excluded

- Honeypot (intentionally isolated)
 - Web Server (optional; not joined during this phase)
-

Step 1: Verify Network Connectivity

Verify the Linux server can communicate with the Domain Controller.

```
ping 192.168.150.40
```

```
ping kpk-dc01.kpk.local
```

Verify DNS:

```
nslookup kpk-dc01.kpk.local
```

```
dig kpk.local
```

```
dig -t SRV _ldap._tcp.dc._msdcs.kpk.local
```

```
dig -t SRV _kerberos._tcp.dc._msdcs.kpk.local
```

Step 2: Configure DNS

Set the Domain Controller as the primary DNS server.

DNS Server

```
192.168.150.40
```

Verify:

```
cat /etc/resolv.conf
```

```
resolvectl status
```

Expected:

```
nameserver 192.168.150.40
```

Step 3: Install Required Packages

Update repositories.

```
sudo apt update
```

Install AD integration packages.

```
sudo apt install realmd sssd sssd-tools libnss-sss libpam-sss adcli samba-common-bin oddjob oddjob-  
mkhomedir packagekit krb5-user -y
```

Verify installation.

```
realm --version
```

```
kinit --version
```

Step 4: Discover the Domain

Verify that Linux can locate the Domain Controller.

```
realm discover kpk.local
```

Expected output includes:

- Domain Name
 - Kerberos Realm
 - Domain Controller
 - Client Software
 - Required Packages
-

Step 5: Join the Domain

Join using the Domain Administrator account.

```
sudo realm join kpk.local -U Administrator
```

Enter the domain password when prompted.

Step 6: Verify Domain Join

Verify membership.

realm list

Expected:

domain-name: kpk.local

configured: kerberos-member

Step 7: Verify Kerberos Authentication

Obtain a Kerberos ticket.

kinit Administrator

Verify:

klist

Expected:

- Valid TGT
 - Realm: KPK.LOCAL
-

Step 8: Verify AD User Resolution

Query an Active Directory user.

id Administrator

getent passwd Administrator

Expected:

- UID
 - GID
 - Home Directory
 - Domain Information
-

Step 9: Configure Automatic Home Directory Creation

Enable automatic home directory creation.

sudo pam-auth-update

Enable:

Create home directory on login

Or:

```
sudo pam-auth-update --enable mkhomedir
```

Restart SSSD.

```
sudo systemctl restart sssd
```

Step 10: Validate SSSD

Check service status.

```
systemctl status sssd
```

Verify cache.

```
sss_cache -E
```

Step 11: Test AD Login

Switch to an AD account.

```
su - Administrator
```

Or connect via SSH.

```
ssh Administrator@localhost
```

Verify:

```
whoami
```

```
hostname
```

```
id
```

Step 12: Verify Enterprise Name Resolution

Confirm that all enterprise systems resolve through Active Directory DNS.

```
nslookup misp.kpk.local
```

```
nslookup soc-core.kpk.local
```

```
nslookup ids.kpk.local
```

nslookup web.kpk.local

nslookup kpk-dc01.kpk.local

Step 13: Repeat on Remaining Servers

Perform the same procedure on:

- SOC-Core
- IDS
- MISP

Each server should:

- Use 192.168.150.40 as DNS.
- Successfully discover kpk.local.
- Join the domain.
- Authenticate using Kerberos.
- Resolve AD users through SSSD.

Phase 3 Deliverables

- Linux servers successfully joined to the kpk.local domain.
- Active Directory discovered using realmd.
- Kerberos authentication operational.
- SSSD configured for centralized identity management.
- AD users resolved on Linux.
- Automatic home directory creation enabled.
- Enterprise DNS resolution verified.
- SOC-Core, IDS, and MISP integrated into the Active Directory environment for centralized authentication.